

Project Name:
FitLife Management System

Group Student Names:

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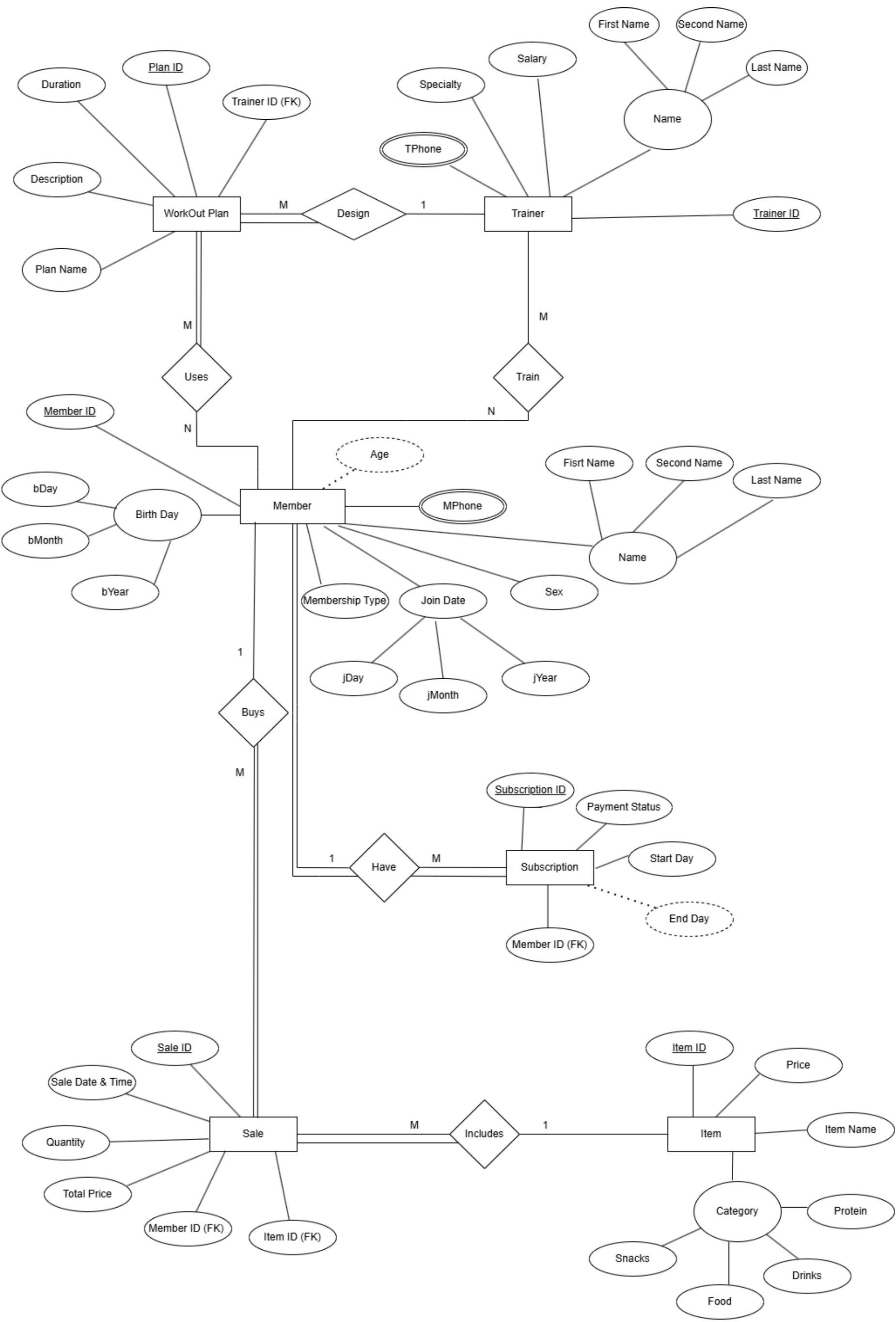
Leader Name:
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Abstract:

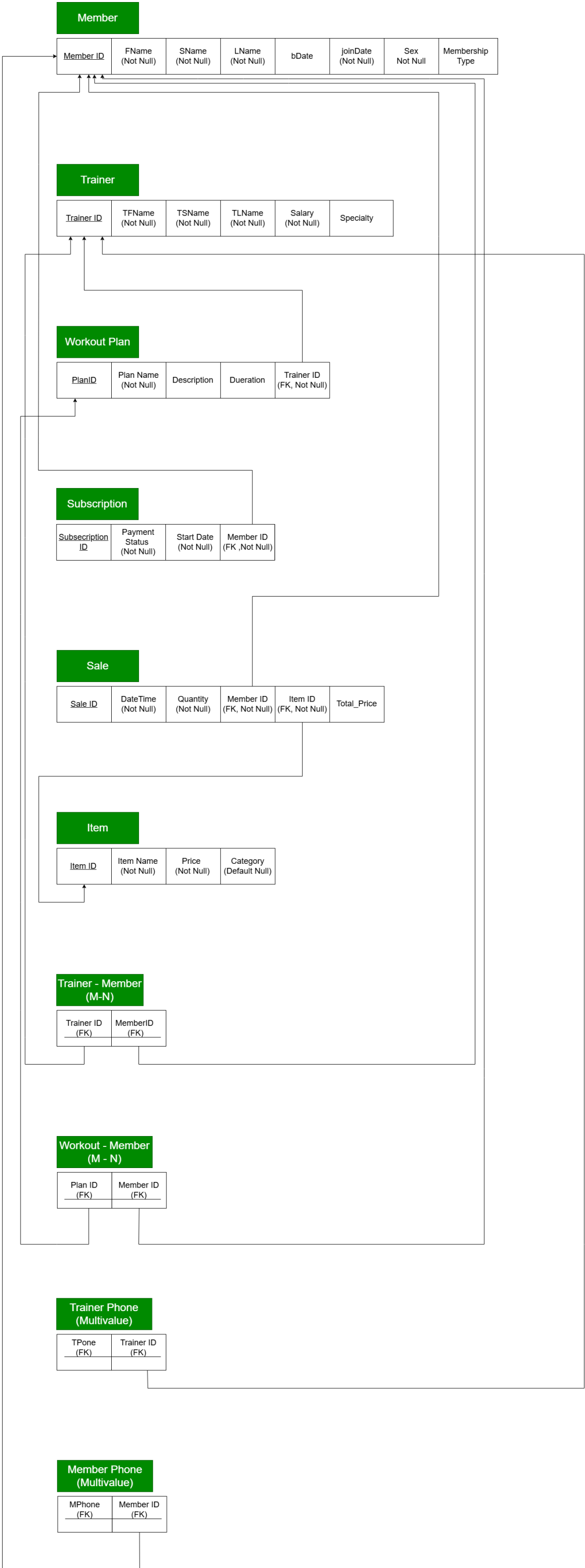
The **FitLife Management System** is a comprehensive database project developed for **Sika Gym** to manage all aspects of gym operations and healthy food services. The system includes several main components such as member registration, trainer management, workout plans, equipment tracking, and maintenance scheduling. It also manages subscriptions, payments, and cafeteria sales of healthy food and protein products. Through this system, Sika Gym can efficiently organize data related to members, trainers, and facilities, while maintaining smooth tracking of daily activities and finances.

The project will be developed for **Sika Gym**, located in **Qalqilya**, and the contact person for this project is **Coach. Ithad Shawahneh**, the gym manager, who can be reached at **0598-470-018**.

ERD (Conceptual Schema)



Mapping (Logical Schema)



Physical Schema / Database Implementation

--Tables Creation

```
create table mem (  
member_id varchar2(20) primary key,  
fname varchar2(50) not null,  
sname varchar2(50) not null,  
lname varchar2(50) not null,  
bDate date,  
jDate date not null,  
sex varchar2(7) not null,  
membership_type varchar2(100),  
check (sex in('m','f','male','female' ))  
);
```

```
create table trainer(  
trainer_id varchar2(20) primary key,  
tfname varchar2(50) not null,  
tsname varchar2(50) not null,  
lname varchar2(50) not null,  
salary number(4) not null,  
specialty varchar2(100),  
check(salary between 1000 AND 8000)  
);
```

```
create table workout_plan(  
plan_id varchar2(20) primary key,  
plan_name varchar2(50) not null,  
plan_description varchar2(200),  
plan_duration varchar2(50),  
trainer_id varchar2(20) not null,  
foreign key (trainer_id) references trainer(trainer_id) on delete cascade  
);
```

```
create table subsc(  
subsc_id varchar2(20) primary key,  
payment_status varchar(10) not null,  
start_date date not null,  
member_id varchar2(20) not null,  
check (payment_status in('paid' ,'unpaid' ,'pending')),  
foreign key (member_id) references mem(member_id) on delete cascade  
);
```

```
create table item(  
item_id varchar2(20) primary key,  
item_name varchar2(50) not null,  
price number(6,2) not null,  
cat varchar2(30) default null,  
check(price between 10 AND 4999.99),  
check (cat in ('Protein' ,'Food' ,'Drinks' ,'Snacks'))  
);
```

```
create table sale(  
sale_id varchar2(20) primary key,  
sale_datetime date not null,  
quantity number (8) not null,  
member_id varchar2(20) not null,  
item_id varchar2(20) not null,  
total_price number(10,2),  
check(quantity between 1 AND 100),  
foreign key (member_id) references mem(member_id) on delete cascade,  
foreign key (item_id) references item(item_id) on delete cascade);
```

```
create table trainer_member(  
-- represents the relation between tranier and member tables ,which is it a M-N Relation  
trainer_id varchar2(20),  
member_id varchar2(20),  
primary key (trainer_id, member_id),  
foreign key (trainer_id) references trainer(trainer_id) on delete cascade,  
foreign key (member_id) references mem(member_id) on delete cascade);
```

```
create table workout_member(  
-- represents the relation between WorkoutPlan and member tables ,which is it a M-N Relation  
plan_id varchar2(20),  
member_id varchar2(20),  
primary key (plan_id,member_id),  
foreign key (plan_id) references workout_plan(plan_id) on delete cascade,  
foreign key (member_id) references mem(member_id) on delete cascade  
);
```

```
create table trainer_phone (  
--multivalued attribute (trainer phone number), a trainer may have more than one phone number  
tphone varchar2(15),  
trainer_id varchar2(20),  
primary key(tphone,trainer_id),  
foreign key (trainer_id) references trainer(trainer_id) on delete cascade);
```

```
create table member_phone(  
--multivalued attribute (member phone number), a member may have more than one phone number  
mphone varchar2(15),  
member_id varchar2(20),  
primary key(mphone,member_id),  
foreign key (member_id) references mem(member_id) on delete cascade);
```


--Data Insertion

--insert data into mem table, aka. Member

insert into mem

values('prm2005', 'Samer', 'Jamal', 'Awayssa',TO_DATE('09-12-2005'),TO_DATE('11-12-2025'),'m',
'Premium');

insert into mem

values('prm465', 'Ibraheem', 'Khalil', 'AbelHaq',TO_DATE('09-12-2005'),TO_DATE('01-10-2022'),'m',
'Premium');

insert into mem

values ('bsc33' , 'Noor' , 'Amroo' , 'Salah',to_date('08-02-2001'),sysdate, 'f' , 'Basic');

insert into mem

values ('gld124' , 'Ali', 'Ahmad', 'Saleh',to_date('12-04-2004'),sysdate, 'm' , 'Gold');

insert into mem

values('slv1002', 'Nabil', 'Yousef', 'Khalil',to_date('22-06-2000'),sysdate, 'm', 'Silver');

insert into mem

values('plt1002', 'Tamer', 'Fadi', 'Rashid',to_date('02-12-1993'),sysdate, 'm', 'Platinum');

--insert data into member_phone table, each member has 2 phone numbers

insert into member_phone

values('+970599567272', 'prm2005');

insert into member_phone

values('+972599567272', 'prm2005');

insert into member_phone

values('+970597597966', 'prm465');

```
insert into member_phone
values('+972597597966', 'prm465');
```

```
insert into member_phone
values('+970598566772', 'bsc33');
```

```
insert into member_phone
values('+972598566772', 'bsc33');
```

```
insert into member_phone
values('+970597456211', 'gld124');
```

```
insert into member_phone
values('+972597456211', 'gld124');
```

```
insert into member_phone
values('+970596453290', 'slv1002');
```

```
insert into member_phone
values('+972596453290', 'slv1002');
```

```
insert into member_phone
values('+970599336081', 'plt1002');
```

```
insert into member_phone
values('+972599336081', 'plt1002');
```

--insert data into trainer table

```
insert into trainer
values('trn5411' , 'Ammar' , 'Ahmad' , 'Salahat' , 8000 , 'Weight Loss');
```

```
insert into trainer
values('trn1102' , 'Mohamed' , 'Hasan' , 'Showpakee' , 4200 , 'Cardio');
```

```
insert into trainer
values('trn2216', 'Omar', 'Fadi', 'Rashid', 6000, 'Strength Training');
```


---insert data into trainer_phone table

insert into trainer_phone
values('+970599111001', 'trn5411');

insert into trainer_phone
values('+972599111001', 'trn5411');

insert into trainer_phone
values('+970599222002', 'trn1102');

insert into trainer_phone
values('+972599222002', 'trn1102');

insert into trainer_phone
values('+970599333003', 'trn2216');

insert into trainer_phone
values('+972599333003', 'trn2216');

--insert data into trainer_member table

insert into trainer_member
values('trn5411', 'prm2005');

insert into trainer_member
values('trn2216', 'prm2005');

insert into trainer_member
values('trn1102', 'prm465');

insert into trainer_member
values('trn5411', 'bsc33');

insert into trainer_member
values('trn2216', 'gld124');

```
insert into trainer_member
values('trn1102', 'slv1002');
```

```
insert into trainer_member
values('trn2216', 'slv1002');
```

```
insert into trainer_member
values('trn5411', 'plt1002');
```

--insert data int subsc table ,aka. subscription table

```
insert into subsc
values('sub001', 'paid', to_date('01-01-2025'), 'prm2005');
```

```
insert into subs
values('sub002', 'pending', to_date('15-09-2025'), 'prm465');
```

```
insert into subsc
values('sub003', 'paid', to_date('01-06-2025'), 'bsc33');
```

```
insert into subsc
values('sub004', 'unpaid', to_date('01-03-2025'), 'gld124');
```

```
insert into subsc
values('sub005', 'paid', to_date('10-07-2025'), 'slv1002');
```

```
insert into subsc
values('sub006', 'paid', to_date('05-02-2025'), 'plt1002');
```

--insert data into workout_plan table

```
insert into workout_plan
values('WP001', 'Cardio Blast', 'High intensity cardio workout', '4 weeks', 'trn5411');

insert into workout_plan
values('WP002', 'Fat Burn Express', 'Quick fat-burning routines', '3 weeks', 'trn5411');

insert into workout_plan
values('WP003', 'Strength Builder', 'Focus on muscle strength and tone', '6 weeks', 'trn1102');

insert into workout_plan
values('WP005', 'CrossFit Power', 'CrossFit training for all levels', '5 weeks', 'trn2216');
```

--insert data into workout_member table

```
insert into workout_member
values('WP003', 'prm465');

insert into workout_member
values('WP001', 'bsc33');

insert into workout_member
values('WP002', 'bsc33');

insert into workout_member
values('WP005', 'gld124');

insert into workout_member
values('WP003', 'slv1002');

insert into workout_member
values('WP005', 'slv1002');

insert into workout_member
values('WP001', 'plt1002');
```

insert into workout_member

values('WP002', 'plt1002');

--insert data into item table

insert into item

values('I001', 'Protein Shake', 35.50, 'Protein');

insert into item

values('I002', 'Whey Protein', 120.00, 'Protein');

insert into item

values('I003', 'Creatine Powder', 55.75, 'Protein');

insert into item

values('I004', 'Protein Bar', 18.99, 'Food');

insert into item

values('I005', 'Chicken Sandwich', 22.50, 'Food');

insert into item

values('I006', 'Energy Drink', 12.00, 'Drinks');

insert into item

values('I007', 'Orange Juice', 10.50, 'Drinks');

insert into item

values('I008', 'Nuts Mix', 12.00, 'Snacks');

insert into item

values('I009', 'Granola Bar', 10.00, 'Snacks');

insert into item

values('I010', 'Tuna Salad', 26.00, 'Food');

--insert data into sale table

insert into sale

values('S001' , SYSDATE, 2, 'prm2005', 'I004', 2* (select price from item where item_id='I004'));

insert into sale

values('S002', SYSDATE, 4, 'bsc33', 'I001', 4 * (select price from item where item_id='I001'));

insert into sale

values('S003', SYSDATE, 4, 'bsc33', 'I003', 4 * (select price from item where item_id='I003'));

insert into sale

values('S004', SYSDATE, 1, 'plt1002', 'I003', 2 * (select price from item where item_id='I003'));

insert into sale

values('S005', SYSDATE, 3, 'prm2005', 'I010', 3* (select price from item where item_id='I010'));

--***Generate some reports***

--View member list

```
select * from mem;
```

--View members with phone numbers

```
select m.member_id, m.fname, m.lname, mp.mphone
from mem m
join member_phone mp on m.member_id= mp.member_id
order by m.member_id;
```

--View trainers with phone numbers

```
select t.trainer_id ,t.tfname, t.lname, tp.tphone
from trainer t
join trainer_phone tp on t.trainer_id=tp.trainer_id
order by t.trainer_id;
```

--View workout plans with trainer name

```
select wp.plan_id, wp.plan_name, wp.plan_description, wp.plan_duration,
t.tfname, t.lname
from workout_plan wp
join trainer t on wp.trainer_id = t.trainer_id;
```


--View sales with member and item details

```
select s.sale_id, s.sale_datetime, s.quantity, s.total_price,  
m.fname, m.lname, i.item_name, i.price  
from sale s  
join mem m on s.member_id= m.member_id  
join item i on s.item_id= i.item_id  
order by s.sale_datetime;
```

--View number of members by membership type

```
select membership_type, count(*) as total_members  
from mem  
group by membership_type;
```

--View total sales per item

```
select i.item_name , sum(s.total_price) as total_sales  
from sale s  
join item i on s.item_id= i.item_id  
group by i.item_name  
order by total_sales desc;
```

--View member with their trainers

```
select m.fname , m.lname,  
t.tfname , t.tlname  
from trainer_member tm  
join mem m on tm.member_id= m.member_id  
join trainer t on tm.trainer_id = t.trainer_id  
order by m.fname;
```

--View members with their workout plans

```
select m.fname, m.lname, wp.plan_name
from workout_member wm
join mem m on wm.member_id = m.member_id
join workout_plan wp on wm.plan_id = wp.plan_id
order by m.fname;
```